

Data Engineering Vol 1

OLAP & DWH

(AWS)

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Thank You Very Much
for the Support !!



Data!!

Finance &
Insurance

Govt.

Data is information.
Collection of facts.
Collection of
measurements.

Predictive
Analytics

AI & ML
(GenAI)

Energy

Marketing

Fraud
Detection

Operational
Analytics

Manufacturing

Retail
E-Commerce

Data

Research

Decision
Support System

Health Care

Transport &
Logistics

Monitoring

Customer
Experience

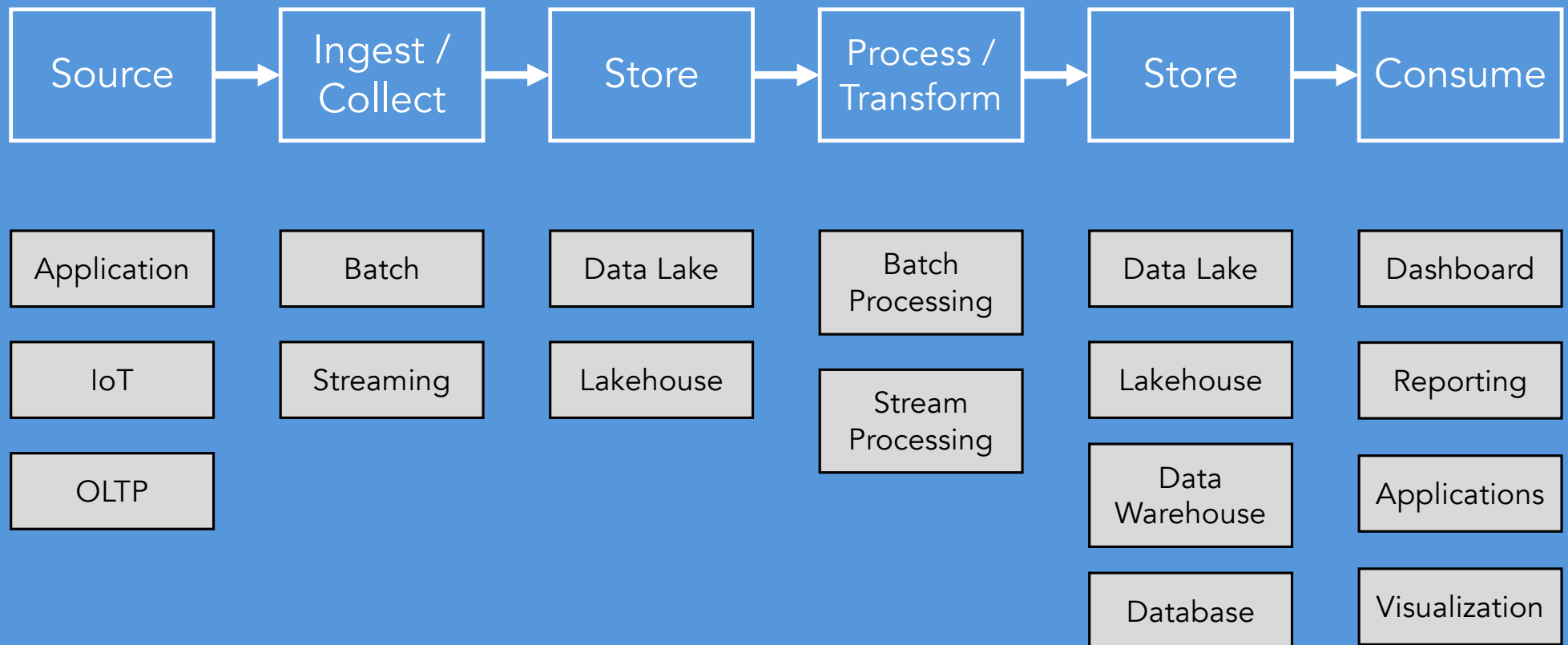
Education

Social Media

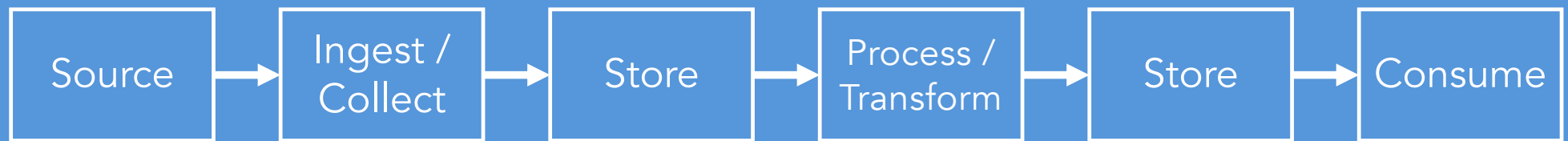
Risk
Management

Autonomous
Driving

Data Life Cycle - Process

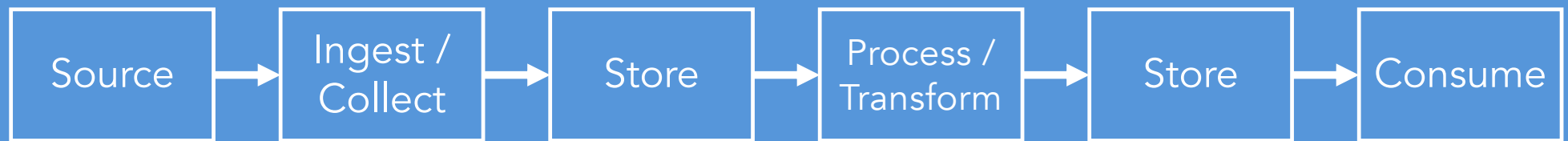


Data Life Cycle - Technologies



Application	Python, Java, C++, Golang, Swift, .NET, C#, C, NodeJS, JavaScript, SQL
IoT	IoT Devices, IoT Core
OLTP	PostgreSQL, MySQL, Oracle, DynamoDB, Cassandra, MongoDB, CouchBase, Redis

Data Life Cycle - Technologies



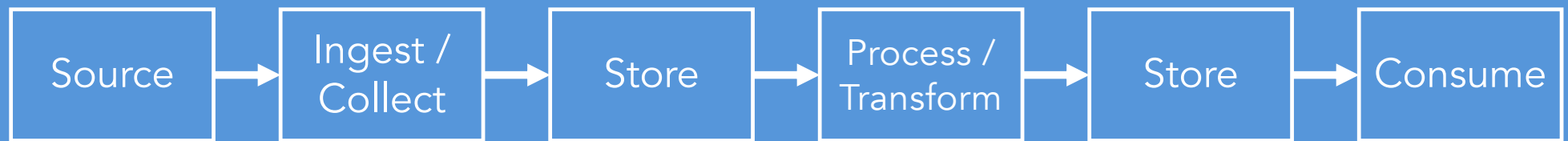
Batch

Python, Java, Spark, Informatica, Talend

Streaming

Kinesis, Kafka, NiFi, Talend

Data Life Cycle - Technologies



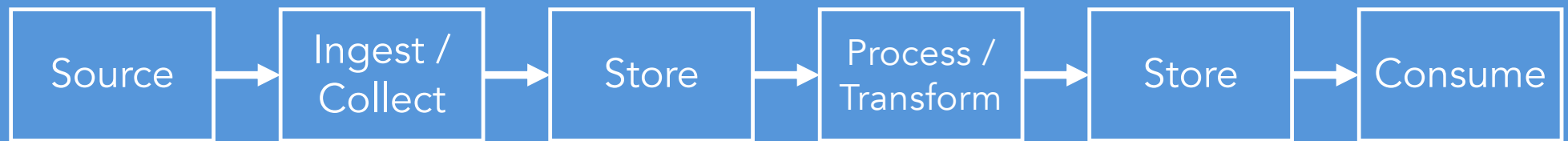
Data Lake

S3, Google Cloud Storage, Azure Data Lake Storage (ADLS)

Lakehouse

HUDI, Iceberg, Delta Lake

Data Life Cycle - Technologies



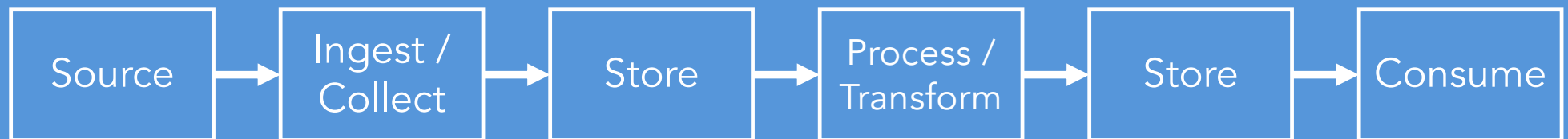
Apache Spark, Apache Hadoop (MapReduce),
Apache Beam

Batch
Processing

Spark Streaming, Apache Flink, Kafka Streams,
Apache Storm

Stream
Processing

Data Life Cycle - Technologies



AWS S3, GCS, Azure Data Lake Storage

Iceberg, HUDI, Delta Lake (Open Table Format)

Redshift, S3, Hive, Snowflake, ClickHouse, BigQuery

RDBMS, NoSQL

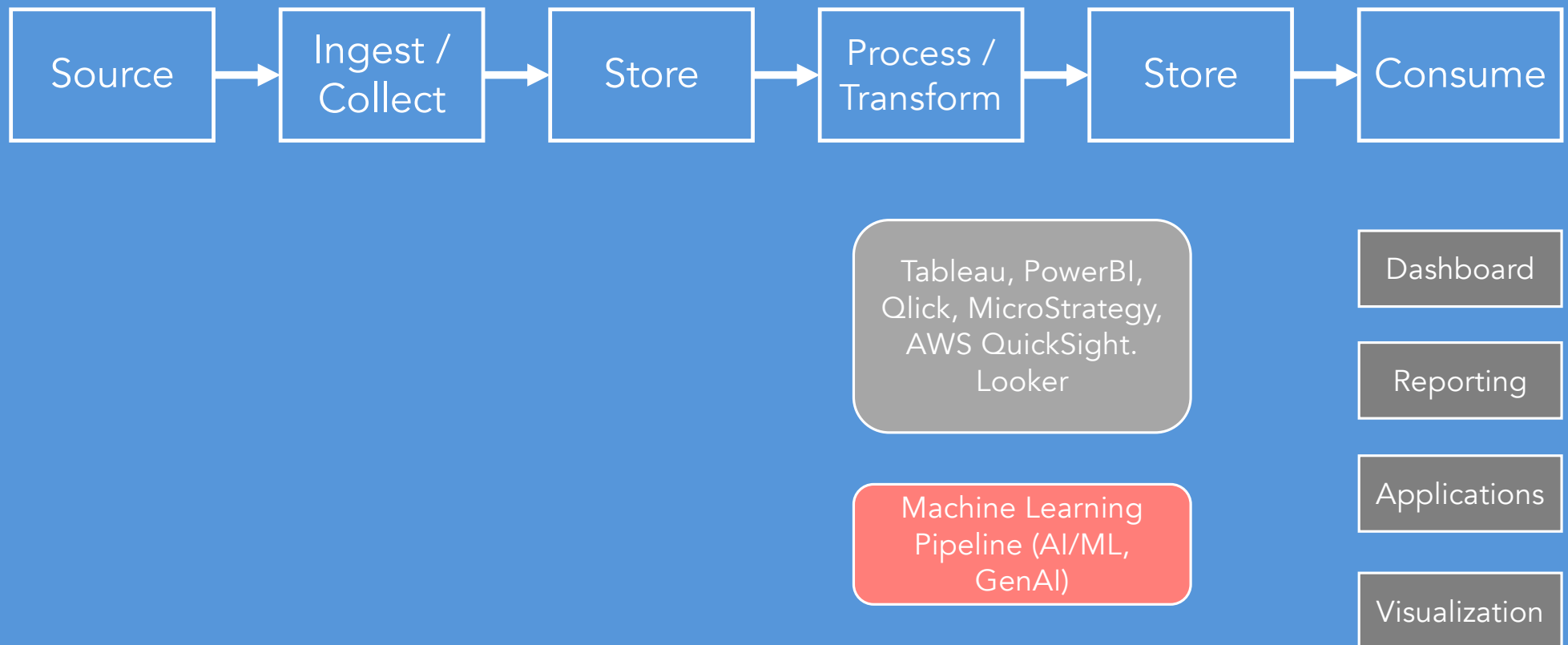
Data Lake

Lakehouse

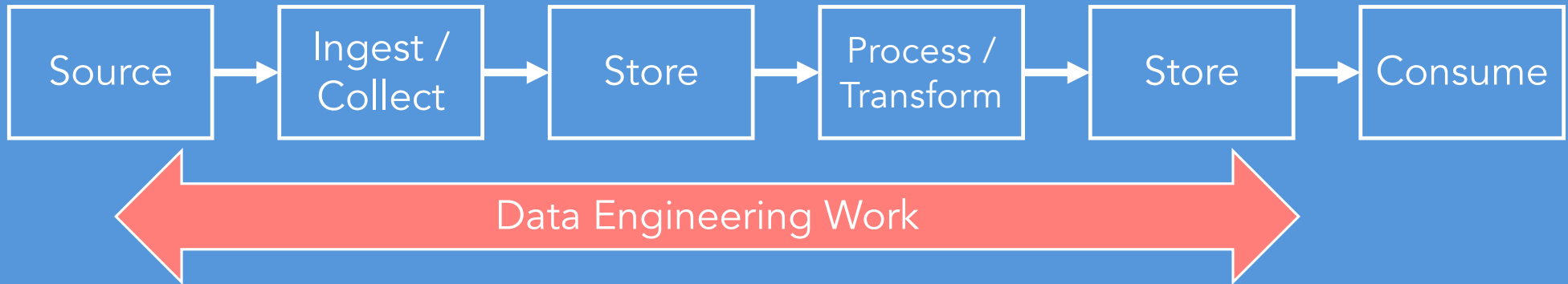
Data Warehouse

Database

Data Life Cycle - Technologies

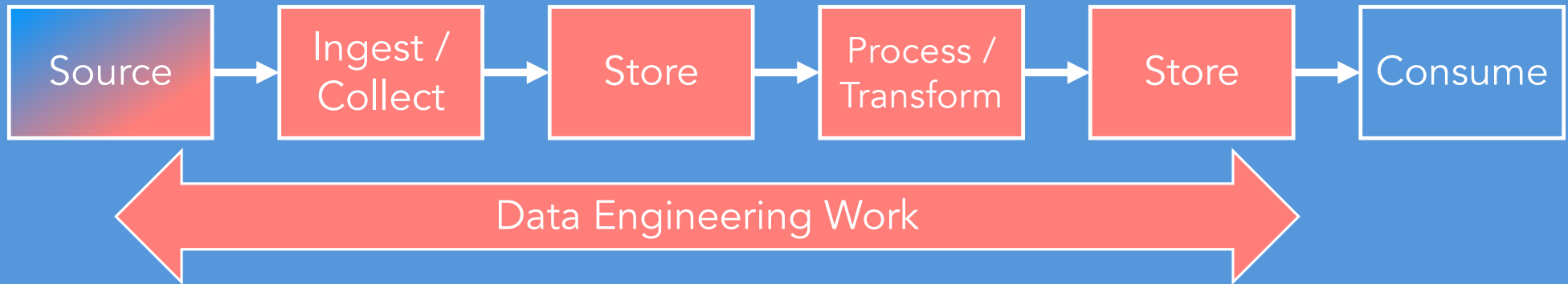


What is Data Engineering



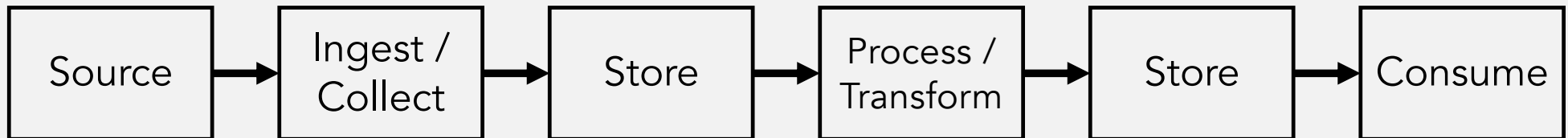
- Data must be engineered for the downstream consumption applications.
- Data Engineering is the process of building **scalable** systems to **ingest, store and analyze data**.
- Data Engineers build the above.

What is Data Engineering



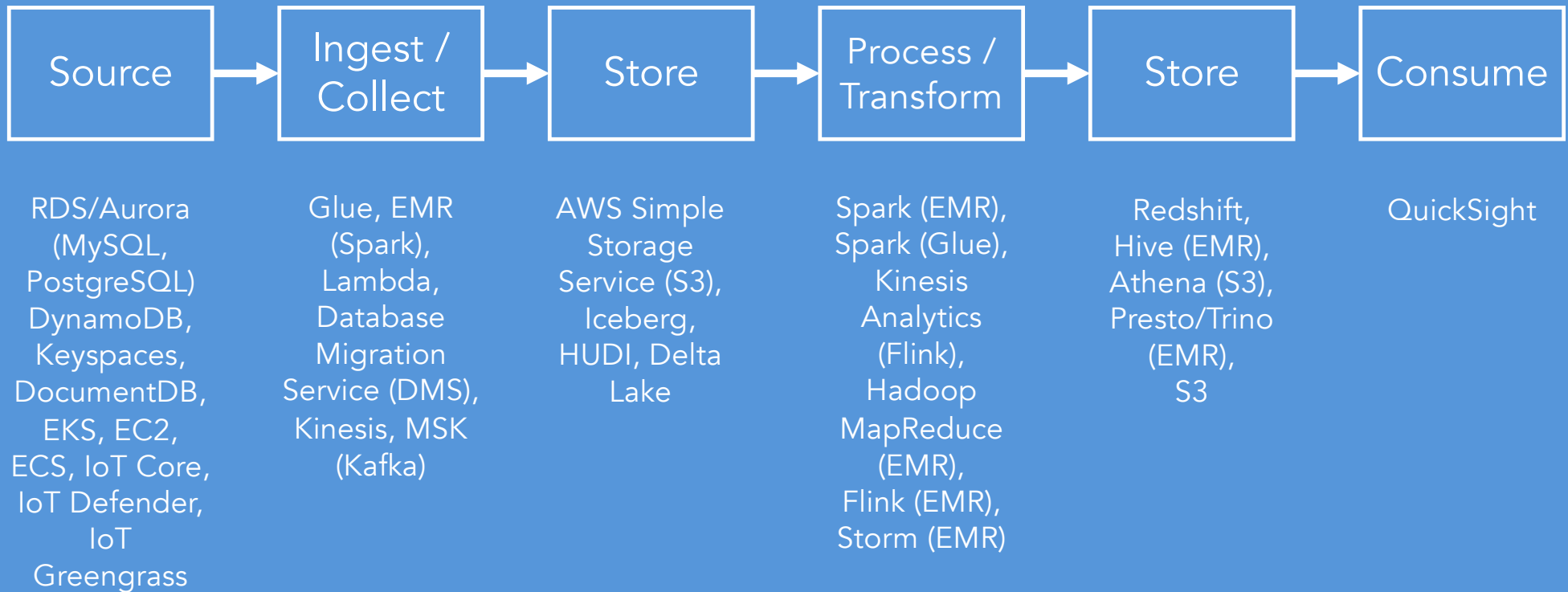
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Different Roles in Data

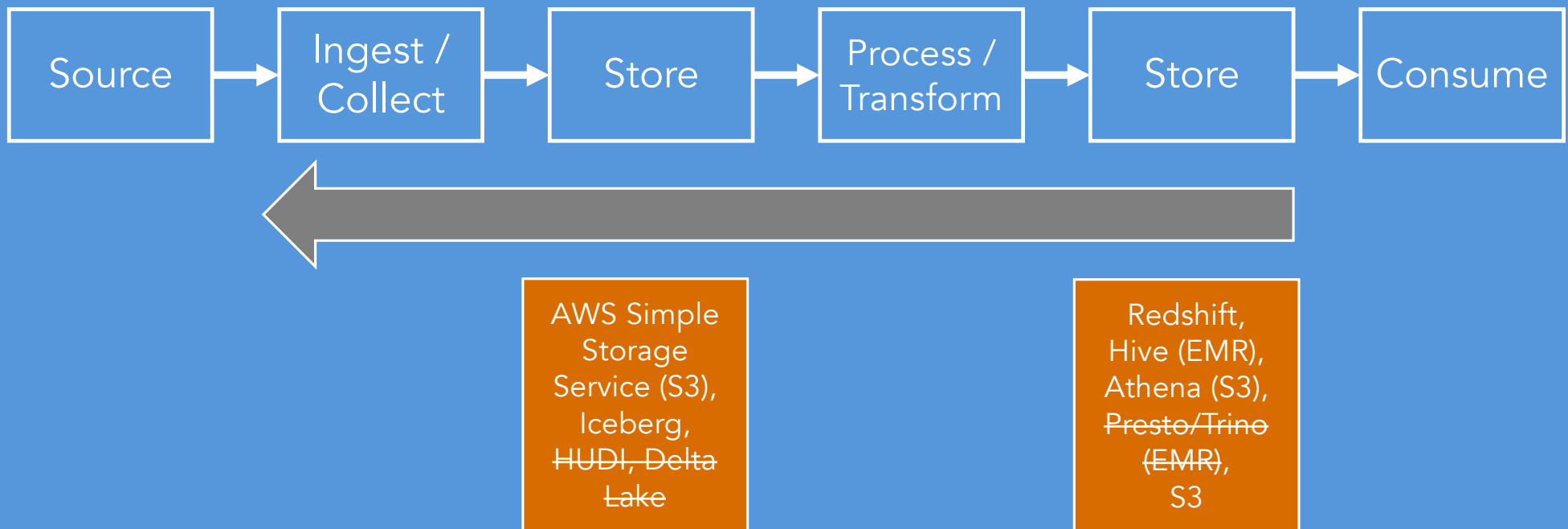


- **Data Engineers** – Ingest or collect, store and analyze data.
- **Data Analyst** – Data interpretation by collecting, preparing, analyzing and visualizing data. **From Store.**
- **Data Scientist** – Need to have mathematics, statistics, ML background. Collect, clean, analyze to find business insights. Data Analyst + ML.
- **Data Architect** – Subset of Data Engineers. Data modelling, data governance, compliance, security.
- **Machine Learning Engineers** – Similar to DE, but for ML pipelines.

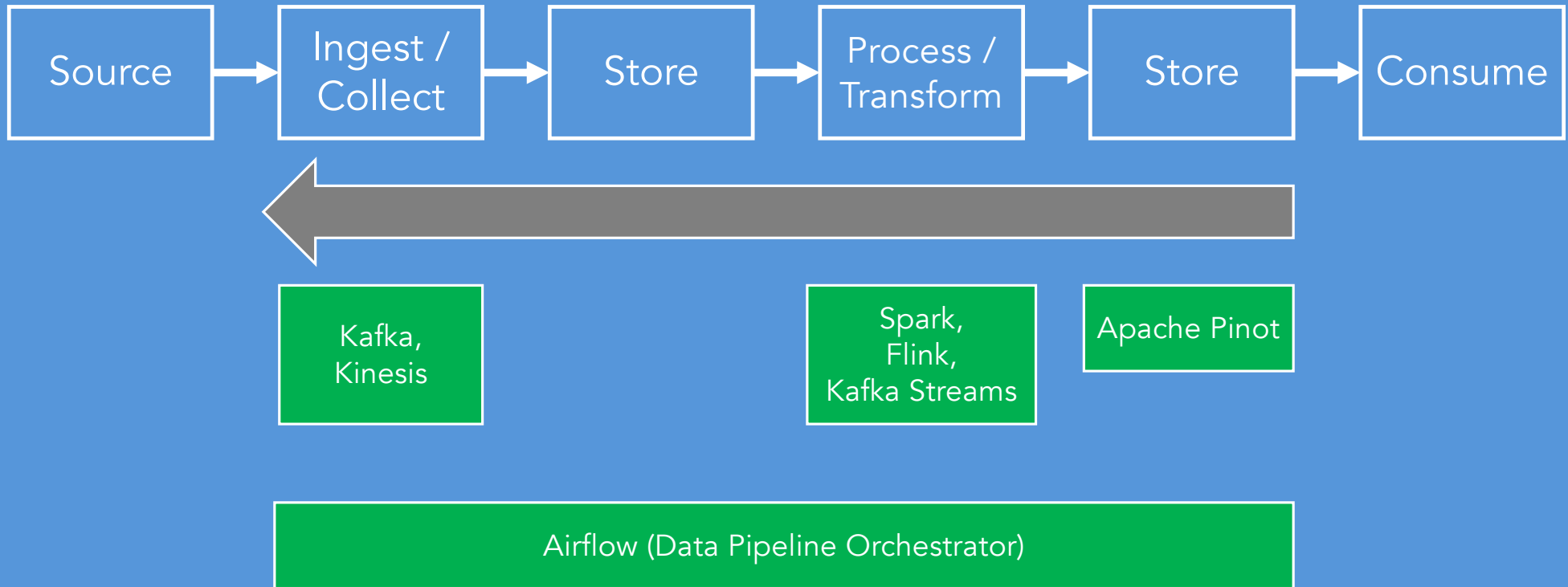
AWS Data Engineering Services



Data Engineering Vol 1 (AWS)



Data Engineering Vol 2



What will be Covered in the Course

AWS Pre-requisites

EC2 (Elastic Compute Cloud)
EBS (Elastic Block Storage)
VPC (Virtual Private Cloud)
IAM (Identity & Access Management)

Main Course

S3 (Simple Storage Service)
Data Modelling (OLTP & OLAP)
Redshift (L100 – L400)
Athena
Hive (on EMR)

Non-AWS Pre-requisites

Python for Data Engineering
SQL (Basic to Advanced)

Course Highlights

- [Live discussion sessions](#) – doubt clearing, use case discussion, technology fitment, career discussion, project, error discussion.
- Hands-on exercise with production like scenarios using [100 to 200 GB of data](#).
- Project assignment (with a different dataset).
- Data Generation ([Up to 2 to 5 GB per dataset](#)).
- Attain Expert level after course completion.



Who is this course for?

- Want to start career on anything that is “Data”.
- Database administrators, Solution architects, Data engineers, Data scientists etc.
- Want to ride the data wave in the industry.
- Want to learn Data Engineering on AWS.
- Want to go deep into AWS services (like Redshift, Athena).
- Want job oriented learning using real life scenarios.
- Want personalize learning experience.
- Help in certification (Redshift, S3, Athena, EMR)

Key Takeaways

- Deep understanding of Redshift, Athena, Hive.
- Understand Data Warehouse, Data Lakehouse, Data Mart and their implementation with Redshift.
- How Redshift and Athena are used for various use cases.
- How to load/ingest data into Redshift for multiple different scenarios.
- Integration of Redshift and Athena with the larger Data Engineering ecosystem.
- Basics of AWS, Python, SQL.
- Can work as Redshift DBA.
- ETC...

Projects (Total Dataset ~ 250 GB)

- Project 1 – Data Modelling and Python exercise. Dataset size – 2 to 5 GB.
- Project 2 – Redshift Queries. Insurance Domain. Dataset size – 35 to 50 GB.
- Project 3 – Redshift Administration. Education Domain. Dataset size – 40 to 50 GB.
- Project 4 – OLTP to OLAP. ELT on Redshift. Dataset size – 15 to 25 GB.
- Project 5 – Redshift Query optimization. Healthcare Domain. Dataset size – 75 to 100 GB.
- Project 6 – Redshift Streaming Ingestion. Order Management. Dataset size – 25 GB.
- Project 7 – Redshift Security. Retail Domain. Dataset size – 1 to 5 GB.
- Project 8 – WLM Implementation using one of the datasets above.
- Project 9 – Athena Queries. Retail Domain. Dataset size – 1 to 5 GB.
- Project 10 – Redshift Transaction, Locking, SUPER data type. Dataset size – 1 to 5 GB.

Let's Begin...

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